Answer key - Simultaneous Equations

GCE Ordinary Level - Mathematics - Corrige

Detailed method for every exercise. Follow the substitutions line by line.
Corrige detaille de chaque exercice. Suis les substitutions ligne par ligne.

1. Solve y = x + 1 and 3x + y = 9.
Résous y = x + 1 et 3x + y = 9.

- x = 2, y = 3 *
- x = 1, y = 2
- x = 3, y = 4
- x = 2, y = 2

Answer: x = 2, y = 3
Substitute: 3x + (x + 1) = 9 -> 4x = 8 -> x = 2, y = 3.
Remplace : 3x + (x + 1) = 9 -> 4x = 8 -> x = 2, y = 3.

2. Solve x + y = 6, x - y = 2 by elimination.
Résous x + y = 6, x - y = 2 par élimination.

- x = 4, y = 2 *
- x = 3, y = 3
- x = 4, y = 1
- x = 2, y = 4

Answer: x = 4, y = 2
Add: 2x = 8 -> x = 4, y = 2.
Additionne : 2x = 8 -> x = 4, y = 2.

3. Solve 2x + y = 10, x + 2y = 8.
Résous 2x + y = 10, x + 2y = 8.

- x = 4, y = 2 *
- x = 2, y = 4
- x = 3, y = 4
- x = 4, y = 3

Answer: x = 4, y = 2
Multiply second by 2: 2x + 4y = 16. Subtract first: 3y = 6 -> y = 2, x = 4.
Multiplie la seconde par 2 : 2x + 4y = 16. Soustrais la première : 3y = 6 -> y = 2, x = 4.

4. Solve y = x + 2, x + y = 8.
Résous y = x + 2, x + y = 8.

- x = 3, y = 5 *
- x = 2, y = 4
- x = 5, y = 7
- x = 4, y = 6

Answer: x = 3, y = 5
x + (x+2)=8 -> 2x=6 -> x=3, y=5.

5. Solve x + y = 7, x - y = 3.
Résous x + y = 7, x - y = 3.

- x = 5, y = 2 *

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- x = 4, y = 3
- x = 3, y = 4
- x = 2, y = 5

Answer: x = 5, y = 2
Add: 2x=10 -> x=5, y=2.
Additionne : 2x=10 -> x=5, y=2.

6. Solve 2x + y = 11, x - y = 1.
Résous 2x + y = 11, x - y = 1.

- x = 4, y = 3 *
- x = 3, y = 2
- x = 5, y = 1
- x = 4, y = 2

Answer: x = 4, y = 3
Add: 3x=12 -> x=4, y=3.
Additionne : 3x=12 -> x=4, y=3.

7. Solve x + y = 5, 2x - y = 1. The value of x is a. Find a.
Résous x + y = 5, 2x - y = 1. La valeur de x est a. Trouve a.

Answer: 2
Add: 3x = 6 -> x = 2, y = 3.
Additionne : 3x = 6 -> x = 2, y = 3.

8. Solve y = 2x + 1, x + y = 10. The value of y is a. Find a.
Résous y = 2x + 1, x + y = 10. La valeur de y est a. Trouve a.

Answer: 7
Substitute: x + 2x + 1 = 10 -> 3x = 9 -> x = 3, y = 7.
Remplace : x + 2x + 1 = 10 -> 3x = 9 -> x = 3, y = 7.

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